
GCVE-BCP-07 - Known Exploited Vulnerability - KEV Assertion Format

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1 Known Exploited Vulnerability - KEV Assertion Format



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2 Introduction

Known Exploited Vulnerabilities (KEVs) have become a critical signal for vulnerability prioritization, operational risk management, and policy-driven remediation. Governments, CSIRTs, and sectoral authorities increasingly rely on KEV lists to mandate patching, trigger incident response, or inform compliance decisions. However, existing KEV publications are largely list-based and opaque, often asserting exploitation without clearly expressing who made the claim, when exploitation was observed versus declared, what type of evidence supports it, where it was seen, or with what level of confidence. As KEV data is increasingly consumed by automated systems and cross-border information-sharing mechanisms, the absence of structured, contextual metadata limits interoperability, trust calibration, and analytical reuse.

This Best Current Practice defines a standardized KEV assertion format that preserves the intentionally simple and binary nature of KEV while adding minimal but essential context. Within the GCVE or other ecosystem, where vulnerabilities may be disclosed and referenced by multiple independent authorities, exploitation claims must be clearly distinguishable from vulnerability identifiers and treated as attributable statements rather than universal truths. The format enables multiple, potentially conflicting assertions to coexist, supports explicit attribution and confidence signaling, and facilitates interoperability with existing vulnerability, CSIRT, and policy ecosystems without turning KEV into full threat intelligence or requiring disclosure of sensitive evidence.

3 Known Exploited Vulnerability - KEV Assertion Format

This format describes a **generic KEV (Known Exploited Vulnerability) assertion format**.

The goal is to express *who claims exploitation, when, based on what, where it was observed, and with which level of confidence*, without turning KEV into full threat intelligence. A KEV assertion is usually very binary and lacking some meta-information. The format adds some information which could better capture details about the exploitation. A majority of the fields are optional except *vulnerability*, *status* and *evidence*. `[]`.*source* which are recommended.

3.1 Format

It's a single JSON object (ECMA 404) per KEV entry. The KEV entry is associated to a vulnerability ID in GCVE ID or any known vulnerability identifier.

3.1.1 Sample

3.1.1.1 Combined KEV Assertion

The JSON file below provides an example of a KEV file referencing a GCVE vulnerability ID.

```
1 {
2   "vulnerability": {
3     "vulnId": "GCVE-0-2025-55182"
4   },
5   "status": {
6     "exploited": true,
7     "status_reason": "confirmed",
8     "status_updated_at": "2025-12-24T10:15:00Z"
9   },
10  "characteristics": {
11    "remote_code_execution": true,
12    "authentication_required": false,
13    "local_access_required": false
```



```
14   },
15   "timestamps": {
16     "first_seen_at": "2025-12-03T10:15:00Z",
17     "asserted_at": "2025-12-05T12:10:11Z",
18     "recorded_at": "2025-12-05T13:15:00Z",
19     "last_seen_at": "2025-12-24T09:42:21Z"
20   },
21   "scope": {
22     "observation_regions": ["Europe", "North America"],
23     "victim_countries": ["LU", "BE", "US", "CA"],
24     "sector": ["Telecoms", "Aerospace"],
25     "asset_exposure": ["internet-facing"],
26     "notes": "Regions reflect observed evidence, not global exclusivity"
27   },
28   "evidence": [
29     {
30       "type": "incident_response",
31       "signal": "confirmed_compromise",
32       "confidence": 0.9,
33       "source": "national-csirt",
34       "details": {
35         "observed_outcome": ["initial-access", "rce"],
36         "detection_basis": ["forensics", "log-analysis"]
37       }
38     },
39     {
40       "type": "honeypot",
41       "signal": "in_the_wild_attempts",
42       "confidence": 0.6,
43       "source": "research-honeynet",
44       "details": {
45         "attempt_volume": "high",
46         "successful_exploitation": false
47       }
48     }
49   ],
50   "references": [
51     {
52       "id": "GCVE-0-2025-55182",
53       "url": "https://vulnerability.circl.lu/vuln/CVE-2025-55182#sightings"
54     }
55   ],
56 }
```

3.1.1.2 CISA KEV in BCP-07 Format

The JSON file below provides an example of a KEV file referencing a CISA KEV assertion.

```
1 {
2   "vulnerability": {
3     "vulnId": "CVE-2020-29583"
4   },
5   "status": {
6     "exploited": true,
7     "status_reason": "confirmed",
8     "status_updated_at": "2021-11-03T00:00:00Z"
9   },
10  "timestamps": {
11    "first_seen_at": "2021-11-03T00:00:00Z",
12    "asserted_at": "2021-11-03T00:00:00Z",
13    "recorded_at": "2026-01-22T05:07:44Z"
14  },
15  "evidence": [
16    {
17      "type": "vendor_report",
18      "signal": "successful_exploitation",
19      "confidence": 0.8,
20      "source": "cisa-kev",
21      "details": {
22        "feed": "CISA Known Exploited Vulnerabilities Catalog",
23        "date_added": "2021-11-03",
24        "due_date": "2022-05-03",
25        "vendorProject": "Zyxe1",
26        "product": "Multiple Products",
27        "vulnerabilityName": "Zyxe1 Multiple Products Use of Hard-Coded
28          Credentials Vulnerability",
29        "knownRansomwareCampaignUse": "Unknown",
30        "cwes": [
31          "CWE-522"
32        ]
33      }
34    ],
35    "references": [
36      {
37        "id": "CVE-2020-29583",
38        "url": "https://www.cisa.gov/known-exploited-vulnerabilities-
39          catalog?search_api_fulltext=CVE-2020-29583"
40      },
41      "scope": {
42        "notes": "KEV entry: Zyxe1 Multiple Products Use of Hard-Coded
43          Credentials Vulnerability | Affected: Zyxe1 / Multiple Products
44          | Description: Zyxe1 firewalls (ATP, USG, VM) and AP Controllers
45          (NXC2500 and NXC5500) contain a use of hard-coded credentials
46          vulnerability in an undocumented account (\"zyfw\") with an
47          unchangeable password. | Required action: Apply updates per
48          vendor instructions. | Due date: 2022-05-03 | Known ransomware
```

```
43     campaign use (KEV): Unknown | Notes (KEV): https://nvd.nist.gov/  
44     vuln/detail/CVE-2020-29583"  
43   }  
44 }
```

3.1.2 Field Description

3.1.2.1 vulnerability Object

Describes the vulnerability being asserted as exploited.

3.1.2.1.1 vulnerability.vulnId

- **Type:** string
- **Required:** yes
- **Description:** GCVE, CVE identifier, GHSA or any identifier of the vulnerability.
- **Example:** "GCVE-0-2025-55182"

3.1.2.1.2 vulnerability.altId

- **Type:** array
- **Required:** no
- **Description:** Alternative identifiers that refer to the same vulnerability, used in addition to `vulnerability.vulnId`.

3.1.2.2 status Object

Represents the current exploitation status.

3.1.2.2.1 status.exploited

- **Type:** boolean
- **Description:** Indicates whether exploitation has been observed or asserted.
- **Semantics:** Does not imply global prevalence or universal exploitability.

3.1.2.2.2 status.status_reason

- **Type:** string (enum)
- **Allowed values:** `confirmed`, `suspected`, `disputed`, `historical`, `unknown`
- **Description:** Rationale behind the exploitation status.

3.1.2.2.3 **status.status_updated_at**

- **Type:** string (RFC3339 datetime)
- **Description:** Timestamp of the last change to the exploitation status in the KEV assertion.

3.1.2.3 **characteristics Object**

Describes high-level technical characteristics of the vulnerability that are relevant to exploitation assessment, without providing exploit details or turning the KEV assertion into full threat intelligence.

These fields describe properties of the vulnerability itself, not necessarily every observed exploitation instance.

3.1.2.3.1 **characteristics.remote_code_execution**

- **Type:** boolean
- **Description:** Indicates whether successful exploitation can result in remote code execution.
- **Notes:** Does not imply exploit reliability or ease of weaponization.

3.1.2.3.2 **characteristics.authentication_required**

- **Type:** boolean
- **Description:** Indicates whether authentication is required to exploit the vulnerability.
- **Notes:** Reflects the weakest known exploitation path.

3.1.2.3.3 **characteristics.local_access_required**

- **Type:** boolean
- **Description:** Indicates whether local system access is required prior to exploitation.
- **Notes:** Useful to distinguish remote exploitation from post-compromise privilege escalation.

3.1.2.3.4 **characteristics.severity**

- **Type:** number (0.0–100)
- **Description:** Severity associated with this vulnerability.

3.1.2.4 **timestamps Object**

Separates different notions of time to avoid ambiguity.

3.1.2.4.1 timestamps.first_seen_at

- **Type:** string (RFC3339 datetime)
- **Description:** Earliest known exploitation activity based on technical observation.
- **Notes:** May be estimated and updated retroactively.

3.1.2.4.2 timestamps.asserted_at

- **Type:** string (RFC3339 datetime)
- **Description:** Date when an authority or source officially declared exploitation.
- **Notes:** Mirrors fields such as “date added” in KEV lists.

3.1.2.4.3 timestamps.recorded_at

- **Type:** string (RFC3339 datetime)
- **Description:** Timestamp when this assertion was ingested or recorded by the collector.
- **Notes:** System-specific and independent of the source.

3.1.2.4.4 timestamps.last_seen_at

- **Type:** string (RFC3339 datetime)
- **Description:** Most recent confirmed observation of exploitation activity.
- **Notes:** Optional and often unavailable.

3.1.2.5 scope Object

Defines the observed context of exploitation.

3.1.2.5.1 scope.observation_regions

- **Type:** array of strings
- **Description:** Geographic regions where exploitation evidence was observed. The region can be described in **UN M49** format to facilitate automation.
- **Notes:** Reflects sensor or reporting coverage, not global limits.

3.1.2.5.2 scope.victim_countries

- **Type:** array of strings
- **Description:** Countries in ISO 3166 where confirmed victims were identified.
- **Notes:** Often incomplete or unavailable.

3.1.2.5.3 scope.sector

- **Type:** array of strings
- **Description:** Sectors targeted or affected by exploitation. The sector **SHALL** come from the **MISP galaxy sector**.
- **Example:** "Telecoms", "Aerospace"

3.1.2.5.4 scope.asset_exposure

- **Type:** array of strings
- **Allowed values:** internet-facing, internal, vpn-accessible, unknown
- **Description:** Exposure context of affected assets.

3.1.2.5.5 scope.notes

- **Type:** string
- **Description:** Human-readable clarifications to prevent misinterpretation.

3.1.2.6 evidence Array

Collection of independent signals supporting the exploitation claim.

3.1.2.6.1 evidence[].type

- **Type:** string (enum)
- **Allowed values:** incident_response, telemetry, honeypot, sinkhole, vendor_report, public_report, research_report, unknown
- **Description:** Origin of the exploitation evidence.

3.1.2.6.2 evidence[].signal

- **Type:** string (enum)
- **Allowed values:** (can be multiple) in_the_wild_attempts, successful_exploitation, confirmed_compromise, mass_scanning, weaponized_exploit_available
- **Description:** Nature of the observed exploitation signal.

3.1.2.6.3 evidence[].confidence

- **Type:** number (0.0–1.0) or enum
- **Description:** Confidence level associated with this evidence.

3.1.2.6.4 `evidence[]`.source

- **Type:** string
- **Description:** Logical identifier of the reporting entity or data source. MISP org UUID? What about existing KEV source like CISA, ENISA or alike. Should we have an enum with existing ones? The source would be the only required fields has many KEV like the type of signal.

3.1.2.6.5 `evidence[]`.details

- **Type:** object
- **Description:** Structured, free-form metadata describing how the signal was derived. Additional feeds from KEV sources which are not described in this format such as [cwes](#).
- **Notes:** Content is implementation-specific.

3.1.2.6.6 `evidence[]`.gcve

- **Type:** object
- **Description:** Structured object describing evidence originating from the GCVE ecosystem.

gcve Object

- `evidence[]`.gcve.vluuid
 - **Type:** string
 - **Description:** UUID of the Vulnerability-Lookup instance where the assertion originated. If the UUID must be derived from a source other than Vulnerability-Lookup, GCVE maintains a list of **known KEVs** to determine the correct source UUID.
- `evidence[]`.gcve.gna
 - **Type:** number (0–65535)
 - **Description:** GNA ID identifying the origin of the assertion.
- `evidence[]`.gcve.object_uuid
 - **Type:** string
 - **Description:** UUID of the assertion associated with this evidence in the GCVE ecosystem.

3.1.3 JSON Schema

JSON Schema - GCVE-BCP-07 Known Exploited Vulnerability (KEV) Assertion Format.

```
1 {
2   "$schema": "https://json-schema.org/draft/2020-12/schema",
3   "$id": "https://gcve.eu/schemas/bcp-07-kev-assertion.schema.json",
4   "title": "GCVE-BCP-07 Known Exploited Vulnerability (KEV) Assertion",
5   "type": "object",
6   "additionalProperties": false,
7   "required": ["vulnerability", "status"],
8   "properties": {
9     "vulnerability": {
10      "type": "object",
11      "additionalProperties": false,
12      "required": ["vulnId"],
13      "properties": {
14        "vulnId": {
15          "type": "string",
16          "description": "GCVE, CVE, GHSA or any identifier of the
17            vulnerability."
18        },
19        "altId": {
20          "type": "array",
21          "description": "Alternative identifiers that refer to the
22            same vulnerability, used in addition to vulnerability.
23            vulnId.",
24          "items": { "type": "string" }
25        }
26      }
27    },
28    "status": {
29      "type": "object",
30      "additionalProperties": false,
31      "properties": {
32        "exploited": {
33          "type": "boolean",
34          "description": "Indicates whether exploitation has been
35            observed or asserted."
36        },
37        "status_reason": {
38          "type": "string",
39          "description": "Rationale behind the exploitation status.",
40          "enum": ["confirmed", "suspected", "disputed", "historical",
41            "unknown"]
42        },
43        "status_updated_at": {
44          "type": "string",
45          "format": "date-time",
46          "description": "Timestamp of the last change to the
47            exploitation status in the KEV assertion (RFC3339)."
```



```
45     },
46
47     "characteristics": {
48         "type": "object",
49         "additionalProperties": false,
50         "description": "High-level technical characteristics relevant to
51             exploitation assessment.",
52         "properties": {
53             "remote_code_execution": {
54                 "type": "boolean",
55                 "description": "Whether successful exploitation can result in
56                     remote code execution."
57             },
58             "authentication_required": {
59                 "type": "boolean",
60                 "description": "Whether authentication is required to exploit
61                     the vulnerability."
62             },
63             "local_access_required": {
64                 "type": "boolean",
65                 "description": "Whether local system access is required prior
66                     to exploitation."
67             },
68             "severity": {
69                 "type": "number",
70                 "minimum": 0.0,
71                 "maximum": 100.0,
72                 "description": "Severity associated with this vulnerability -
73                     (0.0100)."
```

```
74     }
75 }
76 },
77
78 "timestamps": {
79     "type": "object",
80     "additionalProperties": false,
81     "description": "Separate notions of time to avoid ambiguity.",
82     "properties": {
83         "first_seen_at": {
84             "type": "string",
85             "format": "date-time",
86             "description": "Earliest known exploitation activity based on
87                 technical observation (RFC3339)."
```

```
88     },
89     "asserted_at": {
90         "type": "string",
91         "format": "date-time",
92         "description": "Date when an authority or source officially
93             declared exploitation (RFC3339)."
```

```
94     },
95     "recorded_at": {
```

```
89         "type": "string",
90         "format": "date-time",
91         "description": "Timestamp when this assertion was ingested/
                        recorded by the collector (RFC3339).",
92     },
93     "last_seen_at": {
94         "type": "string",
95         "format": "date-time",
96         "description": "Most recent confirmed observation of
                        exploitation activity (RFC3339).",
97     }
98 },
99 },
100
101 "scope": {
102     "type": "object",
103     "additionalProperties": false,
104     "description": "Observed context of exploitation.",
105     "properties": {
106         "observation_regions": {
107             "type": "array",
108             "description": "Geographic regions where exploitation
                            evidence was observed (optionally UN M49).",
109             "items": { "type": "string" }
110         },
111         "victim_countries": {
112             "type": "array",
113             "description": "Countries (ISO 3166) where confirmed victims
                            were identified.",
114             "items": {
115                 "type": "string",
116                 "minLength": 2,
117                 "maxLength": 2
118             }
119         },
120         "sector": {
121             "type": "array",
122             "description": "Sectors targeted/affected (SHALL come from
                            MISP galaxy sector).",
123             "items": { "type": "string" }
124         },
125         "asset_exposure": {
126             "type": "array",
127             "description": "Exposure context of affected assets.",
128             "items": {
129                 "type": "string",
130                 "enum": ["internet-facing", "internal", "vpn-accessible", "
                            unknown"]
131             }
132         },
133         "notes": {
```

```
134         "type": "string",
135         "description": "Human-readable clarifications to prevent
                           misinterpretation."
136     }
137 }
138 },
139
140 "evidence": {
141     "type": "array",
142     "description": "Collection of independent signals supporting the
                           exploitation claim.",
143     "items": { "$ref": "#/$defs/evidenceItem" }
144 },
145
146 "references": {
147     "type": "array",
148     "description": "Links/IDs referencing external resources about
                           the vulnerability or sightings.",
149     "items": { "$ref": "#/$defs/reference" }
150 }
151 },
152
153 "$defs": {
154     "reference": {
155         "type": "object",
156         "additionalProperties": false,
157         "required": ["id", "url"],
158         "properties": {
159             "id": { "type": "string" },
160             "url": { "type": "string", "format": "uri" }
161         }
162     },
163
164     "confidence": {
165         "description": "Confidence level: number -(0.01.0) or an
                           implementation-specific enum/string.",
166         "oneOf": [
167             { "type": "number", "minimum": 0.0, "maximum": 1.0 },
168             { "type": "string" }
169         ]
170     },
171
172     "evidenceSignal": {
173         "oneOf": [
174             {
175                 "type": "string",
176                 "enum": [
177                     "in_the_wild_attempts",
178                     "successful_exploitation",
179                     "confirmed_compromise",
180                     "mass_scanning",
```

```
181         "weaponized_exploit_available"
182     ]
183 },
184 {
185     "type": "array",
186     "items": {
187         "type": "string",
188         "enum": [
189             "in_the_wild_attempts",
190             "successful_exploitation",
191             "confirmed_compromise",
192             "mass_scanning",
193             "weaponized_exploit_available"
194         ]
195     },
196     "minItems": 1,
197     "uniqueItems": true
198 }
199 ]
200 },
201
202 "gcveEvidence": {
203     "type": "object",
204     "additionalProperties": false,
205     "properties": {
206         "vluuid": {
207             "type": "string",
208             "description": "UUID of the Vulnerability-Lookup instance
209                             where the assertion originated."
210         },
211         "gna": {
212             "type": "integer",
213             "minimum": 0,
214             "maximum": 65535,
215             "description": "GNA ID identifying the origin of the
216                             assertion."
217         },
218         "object_uuid": {
219             "type": "string",
220             "description": "UUID of the assertion associated with this
221                             evidence in the GCVE ecosystem."
222         }
223     }
224 },
225
226 "evidenceItem": {
227     "type": "object",
228     "additionalProperties": false,
229     "required": ["source"],
230     "properties": {
231         "type": {
```

```
229         "type": "string",
230         "description": "Origin of the exploitation evidence.",
231         "enum": [
232             "incident_response",
233             "telemetry",
234             "honeypot",
235             "sinkhole",
236             "vendor_report",
237             "csirt_report",
238             "public_report",
239             "research_report",
240             "unknown"
241         ]
242     },
243     "signal": {
244         "$ref": "#/$defs/evidenceSignal",
245         "description": "Nature of the observed exploitation signal (
246             string or array of strings)."
247     },
248     "confidence": { "$ref": "#/$defs/confidence" },
249     "source": {
250         "type": "string",
251         "description": "Logical identifier of the reporting entity or
252             data source."
253     },
254     "details": {
255         "type": "object",
256         "description": "Structured, free-form metadata describing how
257             the signal was derived (implementation-specific).",
258         "additionalProperties": true
259     },
260     "gcve": {
261         "$ref": "#/$defs/gcveEvidence",
262         "description": "Structured object describing evidence
263             originating from the GCVE ecosystem."
264     }
265 }
```

4 References

- `gcve-eu-kev` scripts - CISA KEV and ENISA CNW EUVD to GCVE BCP-07 Converter: <https://github.com/gcve-eu/gcve-eu-kev>

5 Acknowledgements

5.1 BCP-07 Coordinators

- Cédric Bonhomme, CIRCL
- Alexandre Dulaunoy, CIRCL

5.2 Contributions

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- Howard Chu
- Xavier Claude
- Darses
- Jerry Gamblin
- William Robinet

